ATOMIQ QFF 2026 IEM · TECHNICAL SOLUTION DECK

Hybrid Quantum-Classical Term Deposit Campaign Pipeline

Leakage-Free Quantum Kernel Prediction, Constrained 20q QUBO & 10q XY-QAOA Selection, and 6-Qubit Outcome Simulation



Pradyot P. Sahoo | Qiskit 2.6.2 · Qiskit Aer 0.18.2 · Seed = 42

STAGE 1 · PREDICT

Brier 0.0993 · ECE 0.0043

Zero-centered Pauli Kernel (6q)

STAGE 2 · SELECT

23.1835 Obj (+24.40%)

Optimality Gap 0.0000 vs Exact

STAGE 3 · SIMULATE

$P(S \geq 3) = 0.282049$

Mitigated 0.282232 (-95.9% err)

Challenge Aim & Connected 3-Stage Architecture

Core Hackathon Aim (Challenge Specification): Predict subscription propensity for 100 blind test customers (without post-call `duration`), select the optimal **K = 6** outreach cohort from Top-10 candidates maximizing net expected deposit value under budget **C = 25** and advisor capacity **H = 22**, and estimate $P(\text{subscribers} \geq 3)$ via a 6-qubit quantum circuit.

STAGE 1 · PREDICT (6q)

Classical: 5-fold Stratified CV, OHE, `SelectKBest(k=6)`, `tanh` scaling, L2 LogReg, 5 baselines.

Quantum: Zero-centered `Z+ZZ` Pauli feature map $|\phi(x)\rangle$, exact kernel & Aer shots.

→ `stage1_results.csv` **(Top 10)**

STAGE 2 · SELECT (20q / 10q)

Classical: $\binom{10}{6} = 210$ brute-force, exact 2^{20} Ising scan, COBYLA ($\text{CVaR}_{0.15}$).

Quantum: 20q QUBO `CVaR-QAOA` & `SamplingVQE`; 10q $|D^6_{10}\rangle$ XY-QAOA & HW-VEQ.

→ `stage2_results.csv` **(6 IDs)**

STAGE 3 · SIMULATE (6q)

Classical: Exact Poisson-Binomial DP, 100k & 8,192 MC, 2×2 confusion calibration.

Quantum: 6q $R_y(2 \arcsin \sqrt{p_j})$ circuit: Ideal SV, 8,192 shots, Noisy Aer & mitigated.

→ `stage3_results.csv` ($P \geq 3$)

ZERO-LEAKAGE COUPLING

100% Verbatim

Stage 1 probabilities flow untouched into Stages 2 & 3 (enforced to 10^{-12} by unit tests).

OPTIMAL SELECTION UPLIFT

+24.40% Net Value

Cohort **1110001110** scores `23.1835` vs. `18.6361` rank-top-6 (`gap = 0.0000`).

CAMPAIGN TARGET FORECAST

P(S ≥ 3) = 0.282049

Uses 52% budget (13/25; 12 left) & 72.7% cap (16/22; 6 left); mean **1.8970** subs.

Stage 1 Methodology: Dataset Rules, Features & Leakage-Free CV

Dataset Structure & Anti-Leakage Firewall

- **Imbalanced Training Cohort:** `Stage1_Train` has 300 labeled customers (**35** `yes` **= 11.67%**, **265** `no` **= 88.33%**).
- **Post-Call Column Purged:** `duration` is removed before modeling because call duration is unknown before outreach.
- **Strict Single-Shot Scoring:** `Stage1_Test` (100 rows) is untouched during imputation, feature selection, scaling, tuning, and Platt calibration.
- **Deterministic Ranking:** Sorted descending by probability with ascending `customer_id` tie-breaks (`Top 10` pass to Stage 2).

Fold-Isolated Transformation Pipeline (`Seed = 42`)

- **Domain Feature Engineering:** Campaign intensity ratios, previous-contact success flags, macro interaction terms, and age/balance brackets.
- **Per-Fold Preprocessing:** `SimpleImputer`, `OneHotEncoder(handle_unknown='ignore')`, `StandardScaler`, and `SelectKBest(f_classif, k=6)` match the 6-qubit budget strictly inside each training fold.
- **5-Fold Stratified Outer + Inner CV:** Out-of-fold probabilities calibrate via inner Platt scaling ($\sigma(af + b)$) before out-of-sample evaluation.

Why Probability Calibration Is Mission-Critical Here: In Stage 2, net value is $p_i d_i - c_i$; in Stage 3, $P(S \geq 3)$ compounds all six p_j. Mis-calibrated probabilities bias downstream revenue optimization and campaign forecasting even when ranking AUC appears high.

Stage 1 Quantum Core: Fixing Kernel Concentration & Noise Audit

Diagnosing Qiskit Default `zz_feature_map` Collapse

- **Root Cause (Kernel Concentration):** Qiskit's default `zz_feature_map` uses offset phases $2(\pi - x_i)(\pi - x_j)$. Across $k = 6$ standardized features, inner products collapse toward 0 off the diagonal.
- **Symptom Observed:** Off-diagonal Gram entries shrink toward constant noise, degrading 5-fold CV discrimination to near-random (`PR-AUC = 0.1391`, `ROC-AUC = 0.5145`).

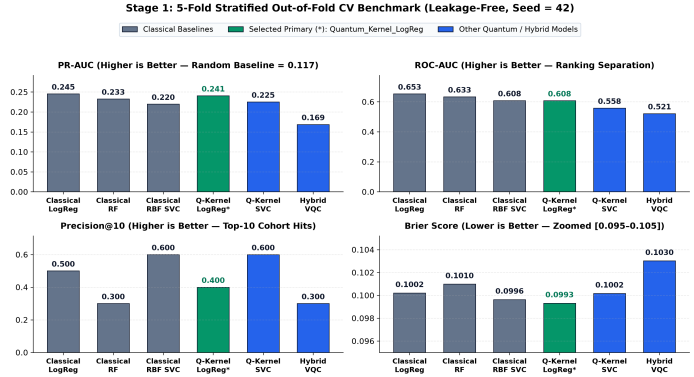
Pauli Kernel + Spectral Whitening

- **Why Zero-Centering Works:** Mapping bounded inputs $z \mapsto b\tanh(z/2)$ into zero-centered `pauli_feature_map(6, r=1, [Z, ZZ])` aligns phases directly with x_i, $x_i x_j$, removing $2\pi^2$ wrap-around.
- **RKHS Spectral Coordinates:** Fidelity matrix $K_{ij} = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$ is whitened inside folds as $F = KV\Lambda^{-1/2}$ for L2 LogReg.
- **Restored Geometry:** Jumps CV to `PR-AUC = 0.2406` (2.06× base rate), `ROC-AUC = 0.6075`, `Brier = 0.0993`, `ECE = 0.0043`.

Aer Kernel Simulation & Noise Audit (12-Customer Subsample, `ComputeUncompute`, 2048 Shots)

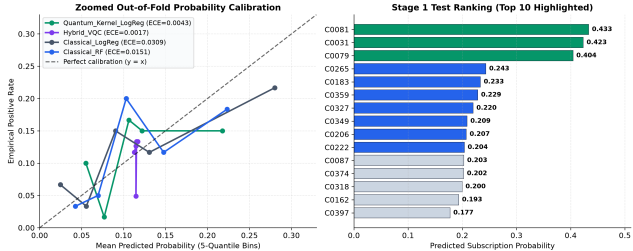
Simulated Shot vs. Noise Match: Finite-shot `AerSimulator` matches exact statevector at **MAE = 0.0052**. Synthetic NISQ errors (0.1% 1Q / 1.0% 2Q depolarizing + readout [0.985/0.015, 0.025/0.975]) raise unmitigated MAE to **0.1574** (`depth = 16`, `10 cx`, `0` SWAPs; simulator audit, not physical QPU).

Stage 1 Benchmark: Interpreting Classical vs. Quantum CV Results



Classical vs. Quantum Summary: **Classical_LogReg** wins top ranking (**PR-AUC = 0.2453**, **ROC-AUC = 0.6527**), and `RBF_SVC`/`QSVC` tie on `Prec@10 = 0.600`. **Quantum_Kernel_LogReg (Green)** is chosen as Primary for **probability calibration**—lowest **Brier (0.0993)**, **ECE = 0.0043** (7× tighter than LR), and `PR-AUC = 0.2406` (2.06× baseline)—not ranking superiority.

Stage 1 Calibration, Test Ranking & 5-Fold Metrics



Out-of-Fold 5-Fold Stratified CV Metrics Across All 6 Classifiers (`Stage1_Train`)

Model	ROC-AUC (OOF ± std)	PR-AUC (OOF ± std)	Prec@10	Rec@10	Brier Score	Log Loss	ECE (5-bin)
`Classical_LogReg`	**0.6527** (±0.126)	**0.2453** (±0.058)	0.500	0.1429	0.1002	0.3584	0.0309
`Classical_RF`	0.6332 (±0.067)	0.2325 (±0.084)	0.300	0.0857	0.1010	0.3500	0.0151
`Classical_RBF_SVC`	0.6083 (±0.060)	0.2195 (±0.117)	**0.600**	**0.1714**	0.0996	**0.3472**	0.0162
`Quantum_Kernel_LogReg` **(Primary)**	**0.6075** (±0.111)	**0.2406** (±0.101)	0.400	0.1143	**0.0993**	0.3484	**0.0043**
`Quantum_Kernel_SVC`	0.5577 (±0.123)	0.2250 (±0.112)	**0.600**	**0.1714**	0.1002	0.3497	0.0089
`Hybrid_VQC`	0.5214 (±0.159)	0.1687 (±0.134)	0.300	0.0857	0.1030	0.3600	**0.0017**

Stage 2 Formulation: 20q QUBO Ising Map & 10q Slack-Free XY-QAOA

Constrained Net-Value Optimization Problem (Ingesting Verbatim Stage 1 p_i)

$$\max_{x\in\{0,1\}^{10}} \sum_{i=0}^{9} (p_i d_i - c_i)\, x_i \quad \text{subject to} \quad \sum_{i=0}^{9} x_i = 6 \text{ (cohort size)}, \quad \sum_{i=0}^{9} c_i x_i \le 25 \text{ (budget)}, \quad \sum_{i=0}^{9} w_i x_i \le 22 \text{ (capacity)}.$$

Path A: 20q QUBO + Exact 2^{20} Ising Check

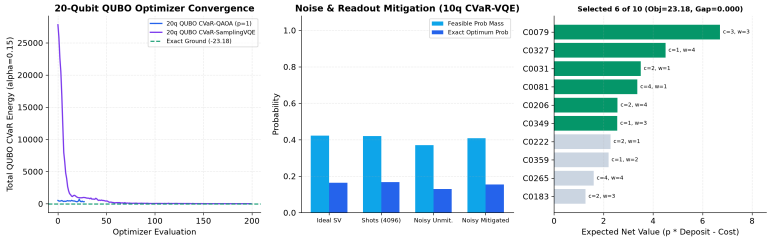
- **Penalty $\lambda = 36.0$:** Exceeds $\sum |v_i|$, placing every invalid bitstring above all valid energies across 10 decision + 10 slack qubits.
- **20q Ising Spectrum:** 185 Pauli Z/ZZ terms (`offset=4484.7345`); exhaustive 2^{20} scan verifies `matches_exact=True` (`-23.183526`).

Path B: Dicke $|D^6_{10}\rangle$ Eliminates Slacks (10q)

- **Subspace XY-QAOA ($p = 2$):** Prepares weight-6 $|D^6_{10}\rangle$ ($\binom{10}{6} = 210$ states); brickwork `RXX(β)RYY(β)` preserves **100.00%** $K = 6$ **feasibility** with 0 slack qubits.
- **10q HW-VQE (Aer Sim):** `RealAmplitudes(r=1)` reaches **16.42%** SV / **16.80%** 4k-shot / **15.42%** mitigated peak on `1110001110`.

Three-Way Optimization Comparison (Exact Reference vs. Penalty vs. Constrained Subspace)

Approach	Qubits / Space	Constraint Mechanism	Feasible Shot Mass	Exact Optimum Peak
1. Classical Brute Force	– / 210 subsets	Direct enumeration	100% exact reference	Deterministic global `1110001110`
2. 20q Penalty QUBO (`QAOA`)	20 / $2^{20} = 1.05$M	10 slacks + penalty $\lambda = 36.0$	Diluted across slacks	`0.1953%` (8/4096 Aer shots, `gap=0.0`)
3. 10q Subspace `XY-QAOA`	10 / 210 Dicke states	Symmetry-preserving $R_{XX}R_{YY}$	**100.00%** feasible	**1.6838%** SV / **1.6357%** shots (8.6×)



All Stage 2 Solvers Reconciled Against Exact Global Optimum (1110001110)

Solver / Simulation Regime	Qubits	Best Bitstring	Best Obj Value	Opt Gap	Feasible Mass / Peak Opt	Key Finding
Classical Brute Force ($\binom{10}{6} = 210$)	–	1110001110	**23.183526**	0.0000	Exact reference	Unique global optimum across all 210 subsets
Exact 2^{20} Ising Diagonal Scan	20	1110001110	**23.183526**	**0.0000**	`matches_exact = True`	Verifies penalty $\lambda = 36.0$, 185 Pauli terms
20q `QUBO_CVaR_QAOA` ($p = 1$, **Aer**)	20	1110001110	**23.183526**	**0.0000**	Peak 0.1953% (8/4096 sh)	Matches global optimum in 28 evals, 0 gap
20q `QUBO_CVaR_SamplingVQE` (Aer)	20	1110001011	22.912935	0.2706	Peak 0.1221% (5/4096 sh)	Finds #2 global solution (200 COBYLA evals)
10q Dicke ($\lvert D^6_{10}\rangle$) `XY-QAOA` (SV / Shots)	10	feasible $K = 6$	CVaR $= -22.37$	–	**100.00%** / 1.68% (1.64%)	Zero slack leakage; 8.62× opt peak vs. QUBO
10q `HW-VQE` (Aer SV / 4,096 Shots)	10	1110001110	**23.183526**	**0.0000**	42.28% (42.07%) / **16.42% (16.80%)**	#1 mode; 0 SWAPs on linear `GenericBackendV2`
10q `HW-VQE` (Aer Noisy → Mitigated)	10	1110001110	**23.183526**	**0.0000**	37.06% → 40.89% / 12.99% → **15.42%**	Simulated readout matrix restores mode fidelity

Stage 2 Cohort Breakdown: Optimal vs. Top-6 (+24.40%)

Top-10 Candidates (`stage2_results.csv`; $C = 25$, $H = 22$, $K = 6$)

Rank / Qubit	Customer ID	Segment	Stage 1 p_i	Cost c_i	Deposit d_i	Cap w_i	Exp. Net $p_i d_i - c_i$	Selected	Portfolio Role
#1 (x0)	**C0081**	C	0.4331	4	17	1	**3.3630**	**Yes (1)**	Core high-propensity anchor
#2 (x1)	**C0031**	D	0.4233	2	13	1	**3.5023**	**Yes (1)**	Core low-cost / high-prob anchor
#3 (x2)	**C0079**	A	0.4043	3	24	3	**6.7034**	**Yes (1)**	Single highest expected net value
#4 (x3)	C0265	D	0.2435	4	23	4	1.5994	Skip (0)	High contact cost ($c = 4$) cuts ROI
#5 (x4)	C0183	B	0.2335	2	14	3	1.2687	Skip (0)	Weak deposit ($d = 14$) yields lowest value
#6 (x5)	C0359	D	0.2285	1	14	2	2.1992	Skip (0)	Outranked on net value by #7–#9
#7 (x6)	**C0327**	B	0.2200	1	**25**	4	**4.5001**	**Yes (1)**	**#2 highest ROI ($d = 25, c = 1$)!**
#8 (x7)	**C0349**	B	0.2089	1	17	3	**2.5507**	**Yes (1)**	Low contact cost ($c = 1$) boosts value
#9 (x8)	**C0206**	A	0.2075	2	22	4	**2.5639**	**Yes (1)**	Large deposit ($d = 22$) outperforms #4–#6
#10 (x9)	C0222	D	0.2038	2	21	1	2.2801	Skip (0)	Strongest Wave-2 reserve candidate

Key Optimization Insight: Picking top-6 propensity (`#1..#6`) returns only `18.6361`. Swapping `#4..#6` (`5.0673`) for high-deposit/low-cost `#7..#9` (`9.6148`) captures **+4.5475 (+24.40% uplift)** = **23.1835 total** using **13/25 budget** and **16/22 capacity**.

Stage 3 Design: 6-Qubit Circuit Mapping & Exact Equivalence

Customer-to-Qubit Mapping (MSB0 Order)

Qubit String: `|q5 q4 q3 q2 q1 q0> = |C0206 C0349 C0327 C0079 C0031 C0081>`

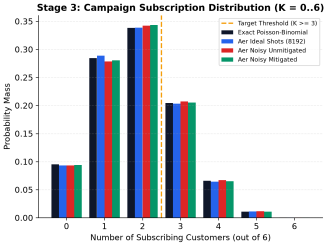
Unentangled $R_y(\theta_j)|0\rangle$, $\theta_j = 2\arcsin\sqrt{p_j}$ so $P(q_j = 1) = p_j$ (depth 2, 0 CNOTs).

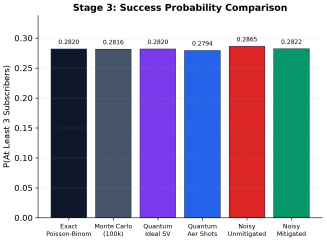
Qubit	String Bit	Customer	S1 Rank	p_j (S1 Verbatim)	$R_y(\theta_j)$ (rad)
q0	b[5]	**C0081**	#1	0.433119	1.436632
q1	b[4]	**C0031**	#2	0.423253	1.416693
q2	b[3]	**C0079**	#3	0.404309	1.378227
q3	b[2]	**C0327**	#7	0.220005	0.976421
q4	b[1]	**C0349**	#8	0.208867	0.949282
q5	b[0]	**C0206**	#9	0.207452	0.945799

Circuit Realization vs. Exact Reference

- **Qiskit Little-Endian Convention:** Bitstring `b = b5 b4 b3 b2 b1 b0` reads qubit j at string index $5 - j$.
- **Success Criterion:** Count shots with Hamming weight $S = \sum_{j=0}^{5} b_j \geq 3$ ($\mathbb{E}[S] = \mathbf{1.8970}$ subscribers).
- **Scope (No Speedup Claim):** Product state $\bigotimes R_y|0\rangle$ is mathematically isomorphic to classical Poisson-Binomial DP (**0.28204944**, $< 10^{-15}$); it provides a circuit-level realization & readout-noise testbed in `AerSimulator`.
- **Tensor Readout Mitigation:** Calibrates 2×2 M_j via $|0^6\rangle$, $|1^6\rangle$ and applies $\bigotimes M_j^{-1}$ to denoise probabilities.

Stage 3 Aer Results: Outcome PMF & 95.9% Error Reduction





Aer 8,192-Shot Counts vs. Exact PMF

Subscribers K	0	1	2	3*	4*	5*	6*
Aer Shot Count	764	2,365	2,774	**1,664**	**528**	**92**	**5**
Aer PMF	0.0933	0.2887	0.3386	0.2031	0.0645	0.0112	0.0006
Exact DP PMF	0.0953	0.2843	0.3384	0.2045	0.0660	0.0108	0.0007

*Success ($K \geq 3$): 2,289/8,192 ideal Aer (0.279419); noisy Aer 2,347/8,192 (0.286499).

Aer Estimator Audit & Why Mitigation Works

Estimator / Simulation Regime	$P(S \geq 3)$	Error vs. Exact
Exact Poisson-Binomial / Ideal SV	**0.282049**	**0.000000**
Classical MC (100,000 / 8,192)	0.281620 / 0.277466	-0.000429 / -0.004584
Ideal Aer Shots (8,192 shots)	0.279419	-0.002630
Noisy Aer (Unmitigated)	0.286499	+0.004450
Noisy Aer + Readout Mitigated	**0.282232**	**+0.000182 (-95.9%)**

Why Mitigation Works: $0 \to 1$ flips (1.5%) promote abundant $K = 2$ shots (33.8%) past $K \geq 3$; tensor inversion $\bigotimes M_q^{-1}$ removes this bias.

Quantum Circuit Telemetry, Seeds & Runtime Verification

Compiled Circuit Telemetry Across All Stages (Aer & Linear `GenericBackendV2`)

Circuit / Backend Target	Logical / Phys.	Orig Depth	Transp. Depth	1Q (Orig → Tr.)	2Q Gates	SWAPs
Stage 1: Zero-Centered Pauli FM (`Z+ZZ`, `r=1`, Aer)	6 / 6	17	16	17 → 11	10 (`cx`)	0
Stage 1: Hybrid VQC (`PauliFM + RealAmp r=2`, Aer)	6 / 6	24	23	35 → 29	20 (`cx`)	0
Stage 2: 20q QUBO `CVaR-QAOA` ($p = 1$, Aer SV)	20 / 20	36	35	60 → 40	165 (`rzz`)	0
Stage 2: 20q QUBO `SamplingVQE` (`RealAmp r=1`, Aer SV)	20 / 20	21	22	40 → 40	19 (`cx`)	0
Stage 2: 10q HW-VQE Ansatz (Linear `GenericBackendV2`)	10 / 10	12	18	20 → 80	9 (`cx`)	0
Stage 3: 6q R_y Circuit (Aer / Linear `GenericBackendV2`)	6 / 6	2	2 / 5	6 → 6 / 24	0	0

Reproducible Stack & Pinned Seeds

- **All 7 Seeds = 42:** `global`, `cv_split`, `model`, `optimizer`, `simulator`, `transpiler`, and `monte_carlo`.
- **Isolated `.venv`:** Python `3.12.11`, `qiskit 2.6.2`, `qiskit-aer 0.18.2`, `qiskit-optimization 0.7.0`, `qiskit-ml 0.9.1`, `scikit-learn 1.9.1`.

Deterministic Execution & Automated Tests

- **End-to-End Runtime:** Stage 1 `17.23s`, Stage 2 `82.46s`, Stage 3 `0.28s` (**104.00s total**, including plots & notebook build).
- `tests/test_pipeline.py` (3/3 tests) & `end_to_end_pipeline.ipynb` executed end-to-end with zero errors.

Submission Evidence Matrix: Verified Repo Deliverables

Evidence-Backed Audit Across Deliverable Artifacts (Stages 1–3 & Tests)

Deliverable / Scope	Required Check	Verified?	Concrete Repository Evidence
`stage1_results.csv`	100 blind IDs, no `duration`	**Verified**	101 lines (100 rows + header); `duration` absent; sorted Top-10 flag
Anti-Leakage & CV	Fold-isolated + classical/quantum	**Verified**	`stage1_cv_metrics.json`: 6 models; `Quantum_Kernel_LogReg` Brier 0.0993, ECE 0.0043
`stage2_results.csv`	$K = 6$, $C \leq 25$, $H \leq 22$, exact ref	**Verified**	11 lines; 1110001110: cost 13/25, cap 16/22, net 23.183526, gap=0.0
Stage 2 Quantum Runs	20q QUBO + 10q XY-QAOA (Aer)	**Verified**	`stage2_summary.json`: 185 Pauli 2^{20} exact match; QAOA 8/4096; XY 100% feas.
`stage3_results.csv`	6q R_y, MSB0, DP/MC/Mitigated	**Verified**	7 lines; 8192 shots; SV 0.282049 = DP; mitigated 0.282232 (-95.9%)
`tests/test_pipeline.py`	$< 10^{-12}$ handoff & schemas	**Verified**	3/3 unit tests pass; `pipeline_metadata.json` logs 7 seeds = 42
Code, Notebook & Docs	Runnable pipeline, README, PDFs	**Verified**	`run_pipeline.py`, executed `.ipynb`, `README.md`, `report.pdf`, `slides.pdf`

Audit Evidence Summary: All CSV schemas, $< 10^{-12}$ cross-stage probability handoffs, exact mathematical references, Aer/transpilation logs, and documentation files are verified in the repository (simulation-based quantum evaluations, not physical QPU runs).

Error Analysis, Limitations & Executive Business Recommendation

RECOMMENDATION

Technical Limitations & Simulator Scope

- **Simulation-Only Quantum Runs:** All experiments execute on `Qiskit Aer` (statevector, shots, synthetic depolarizing/readout noise) + linear `GenericBackendV2` transpilation, not physical QPUs.
- **No Ranking or Stage 3 Speedup Claimed:** `Classical_LogReg` wins Stage 1 `PR-AUC` (`0.2453`) / `ROC-AUC` (`0.6527`)—quantum kernel wins calibration (`ECE=0.0043`). Stage 3 6q R_y is equivalent to Poisson-Binomial DP (noise/mitigation study).
- **QUBO Slacks vs. Scale:** 20q slacks dilute peak shot mass (`0.195%`); 10q XY-QAOA enforces `100%` $K = 6$, and unentangled Stage 3 can extend via `CRy` for correlated customer shocks.

Executive Business Recommendation

- **Deploy Optimal Cohort** `1110001110`**:** Contact **C0081, C0031, C0079, C0327, C0349, C0206** to capture **23.1835 expected net value** (**+24.40%** uplift vs. propensity top-6).
- **Campaign Conversion Guidance:** Expect **1.90 subscribers**; `1-2` conversions occur with **62.3%** probability and the ≥ 3 target with **28.2%** probability (`P = 0.282049`).
- **Wave-2 Growth Headroom:** Uses **52%** budget (`13/25`, **12** left) & **72.7%** capacity (`16/22`, **6** left), enabling Wave-2 follow-up starting with `C0222` (net `2.2801`).

Prepared by: Pradyot P. Sahoo · Full code, CSVs & notebook at `github.com/pradyotqc/Atomiq_QFF2026_IEM`